



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Subject Name	Theory of Computations	A.Y	2026-27
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Assignment-3

Sr No	Question	COs	B.T	Competence
1	Define Context-Free Grammar (CFG) and Context-Free Language (CFL) with suitable examples. Explain their applications in programming language design and syntax analysis.	1	1	Remember
2	Explain Chomsky Normal Form (CNF) with suitable examples. Convert the following CFG into CNF: $S \rightarrow aSB \mid b$ $B \rightarrow b$	3	2	Understand
3	Construct a Pushdown Automaton (PDA) for the language $L = \{a^n b^n \mid n \geq 1\}$. Demonstrate the working of the PDA for the input string “aaabbbb”.	3	3	Apply
4	Analyze ambiguity in Context-Free Grammars using suitable examples. Draw parse trees for the expression $a+a*a$ and determine whether the grammar is ambiguous or unambiguous.	3	4	Analyze
5	Evaluate whether the language $L = \{a^n b^n c^n \mid n \geq 1\}$ is context-free or not using the Pumping Lemma for Context-Free Languages. Justify your answer clearly.	5	5	Evaluate
6	Design a Context-Sensitive Grammar (CSG) for a suitable language and explain how Context-Sensitive Languages differ from Context-Free Languages in terms of computational power and machine acceptance.	4	6	Create